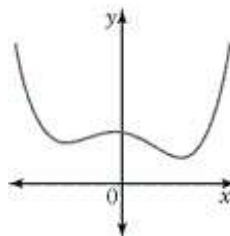


1.3 Graphs of Polynomial Functions-Class Notes

Part I-Multiple Choice

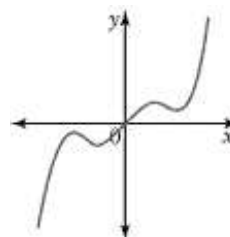
1. The least possible degree of the polynomial function represented by the graph shown is

- a) 2 ☒ c) 4
b) 3 d) 5



2. The **least** possible degree of the polynomial function represented by the graph shown is

- a) 3 ☒ c) 5
b) 4 d) 7



Part I-Multiple Choice

3. If $y = f(x)$ is a quartic function with a constant difference of -48, then the following statement is **false**:

- a) the function starts in Q3 and ends in Q4
b) the sign of the leading coefficient is negative
c) the function might not have any roots
☒ d) the sign of the leading coefficient is positive

1.3 Graphs of Polynomial Functions-Class Notes

Part II- True/False

- a) The function $y = -3x^4 + 1$ extends from quadrant 3 to quadrant 4.

True

- b) Odd-degree polynomials have at least one x-intercept.

True

- c) Even-degree polynomial functions always begin and end on the same side of the x-axis.

True

- d) The graph of a quartic function cannot have exactly three x-intercepts.

False

- e) The function $y = x^4 + 2x^2 + 1$ never crosses the x-axis.

True

- f) All quartic polynomial equations have at least one real solution.

False

Part III-Short Answer

The following is the graph of $y = f(x)$. Answer the following questions.

a) absolute max: 4

b) absolute min: -8

c) local max: 4 & 1

d) local min.: 0

e) interval(s) of increasing : $(-8, -5)$ $(0, 3)$

f) interval(s) of decreasing : $(-5, 0)$ $(3, 6)$

